

Highlights

Local skill, unstable portability: sign reversal in pre-fire thermal predictors across Mediterranean wildfire regions, and the diagnostics that do not order it

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- Pre-fire thermal predictors add +0.06 to +0.15 AUC within all five fire regions
- The same block adds +0.004 AUC across 20 transfer directions, interval spans zero
- Highest niche-overlap pair fails both ways; lowest transfers both ways (points)
- Of 20 diagnostics, only conditional direction agreement tracks transfer (rho 0.84)
- Label-blind adaptation compresses transfer toward chance; only target labels help

Local skill, unstable portability: sign reversal in pre-fire thermal predictors across Mediterranean wildfire regions, and the diagnostics that do not order it

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Abstract

Pre-fire thermal dryness separates a fire year from a normal year, and models built on it are rarely tested outside the region they were fitted in. Five Mediterranean wildfire regions were analysed on about 500 m cells, with MCD64A1 labels and spatially blocked validation, adding six pre-fire thermal predictors to a terrain, fuel and greenness baseline.

The local skill does not travel, and neither does the baseline. Within-region ROC-AUC rose by +0.056 to +0.153 in every region. Across twenty ordered transfer directions the same block contributed +0.004 [−0.028, +0.036], indistinguishable from zero, with a sign that changes from pair to pair. The static baseline transferred no better, at 0.537 against 0.541. Removing the two predictors whose association reverses, elevation and the LST anomaly, cost −0.081 of within-region skill, supported in every region and mostly attributable to elevation, and changed transfer by +0.014 [−0.017, +0.045]. Both are post-selection estimates. A local cost is measured; no compensating transfer gain is.

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The failure is invisible to the diagnostics available before deployment. Of twenty candidate transferability diagnostics rank-correlated against observed transfer, only two had intervals excluding zero and both were conditional, the stronger being agreement in the sign of each predictor’s association (Spearman $\rho = 0.84$ over sixteen directions). No marginal measure was shown to order the matrix, including predictor-space dissimilarity, and at the point estimates the pair with the highest burned-niche overlap failed in both directions while the lowest transferred in both. Signed associations need labels on both sides, so the family that works is the one a practitioner does not have.

The mechanism is a reversal of sign. Predictors do not weaken across regions, they reverse the direction of their association with burning, which is why a distance cannot see the failure. The same reversal appears inside one study area, between two fires eleven months apart on an identical grid; that arm rests on one fire and eleven positive-carrying blocks, so it corroborates rather than establishes.

Label-free alignment pushed fourteen of twenty directions towards chance rather than repairing them. Transfer skill has to be measured rather than inferred from similarity, and the price is target labels: thirty-two labelled 5 km blocks recovered 85 to 89 % of the target’s matched ceiling in three of six directions.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [27]. The pressure to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from both a methodological and an application perspective [39, 13]. The dominant pattern is well established. Geospatial predictors are assembled over a study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [4]. The resulting susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a cost that the pattern does not price. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They are also less portable, as we argue and then show. The relationship they carry is reparameterised locally. It exists everywhere, but it points in different directions in different places. The claim advanced here is a trade-off and not a dismissal: **dynamic state predictors buy local skill at the cost of portability.** Two measurements carry that claim. The dynamic block is worth +0.056 to +0.153 ROC-AUC inside every region and contributes nothing distinguishable from zero between them, with an unstable sign. Removing the reversing predictors costs -0.081 of within-region skill with interval support in every region, while changing mean transfer by +0.014, an estimate whose own interval spans zero. Whether the block helps or harms on transfer is a property of the source-target pair, not of the block.

1.1. Portability goes unmeasured, so the trade-off goes unpriced

A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area and season, often from the same fire event, and where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The problem is documented across ecological modelling [31, 29] and is addressed by spatially blocked cross-validation [37, 23]. Blocking makes the within-region estimate honest. It says nothing about whether the fitted relationship holds elsewhere.

Because only that side of the ledger is reported, the trade-off is structurally invisible. A predictor block is adopted on the strength of the increment it delivers inside its training footprint, and the portability it consumes is never entered as a debit, even though any regional product built from locally trained models implicitly promises generalisation beyond that footprint. For fire specifically the transfer question is asked far less often than in adjacent fields. Meteorological fire-danger indices are known not to port cleanly between fire environments [30], and the two recent studies that test model transfer systematically, across Mediterranean countries [7] and US counties [15], both report that it largely succeeds between similar regions. Both transfer models whose dominant predictors are *spatially stationary*: terrain, human modification, night-time lights, population density, long-term climatologies. Each describes a place rather than a season, which is the class our thesis expects to travel well. Species distribution modelling has examined transferability far more systematically [43], and there geographic and environmental similarity do *not* reliably predict transfer success [38, 32].

1.2. Pre-fire thermal dryness is the natural test case

The dynamic class used here is pre-fire thermal dryness, and it is the class most plausibly *expected* to transfer. Standard susceptibility predictors are terrain, fuel type, greenness, long-term climatology and proximity to roads and settlements, and most are static or near-static over the timescale at which fire danger varies. Such a set explains poorly why a particular summer burned and the preceding one did not. What changes between those summers is the state of the surface: fuel dryness, moisture stress, and how anomalously hot the land surface has become relative to its own baseline. Satellite thermal observation gives direct access to part of that state through fuel moisture content [44], and pairing surface temperature with a vegetation index was established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [6]. The Temperature-Vegetation Dryness Index [33] formalises that feature space into an internally normalised measure which should, in principle, be less exposed to absolute-temperature offsets between regions than raw land surface temperature. That theoretical portability advantage is tested here.

That pre-fire thermal state carries genuine information about subsequent fire is established [17, 19, 18, 8]. What has not been established is whether the predictive *skill* of a classifier trained on it in one fire region survives strict, label-free application to another. The physics linking moisture stress to combustion is universal, so portability should be most expected for this class, which is what makes it diagnostic: a loss here cannot be dismissed as a peculiarity of a locally defined covariate.

75 *1.3. Why the loss is invisible to the diagnostics in use*

A deficit on transfer can arise two ways. Under **covariate shift** the marginal distribution of the predictors differs while the predictor-response relationship is preserved, which is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL
80 [35], an approach with an established remote-sensing literature [36]. Under **concept shift** the conditional relationship itself changes: a predictor positively associated with burning in one region is negatively associated with it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the
85 information being withheld. The taxonomy is canonical [25].

This distinction is what makes the failure hard to anticipate. The diagnostics the field relies on to decide whether a model may be applied to new ground, above all area-of-applicability measures built on dissimilarity in predictor space [21, 22, 16], are marginal by construction. A target region
90 can sit well inside the training data’s predictor envelope while the relationship between those predictors and burning points the other way. This paper measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper. Each is stated at the strength its interval
95 supports, and the nearest prior work is named inside the claim rather than omitted.

Contribution 1. Local skill and portability, measured separately and per direction. For every ordered region pair, the change in transfer skill from adding the dynamic pre-fire thermal block to a matched baseline is

100 set against the within-region increment the same block delivers. The block is
 worth +0.056 to +0.153 ROC-AUC inside every one of five regions, with every
 bootstrap interval above zero and the result holding in both analysis popu-
 lations. Across the twenty ordered transfer directions it contributes +0.004,
 an estimate whose interval spans zero under all four between-direction re-
 105 sampling units we computed, and its sign is unstable: paired deltas run
 from -0.148 to $+0.132$, twelve positive and eight negative. The control that
 bounds the interpretation is the baseline arm: it transfers at a mean of
 0.537 against the thermal model's 0.541 , so the static predictor class is not
 the portable one here either, and what the matrix shows is that **nothing**
 110 **in this feature space travels** rather than that dynamic state is uniquely
 fragile. Feature removal measures the local cost directly. Dropping the two
 reversing predictors, elevation and the LST anomaly, costs -0.081 of mean
 within-region AUC, supported in every region and roughly three quarters of
 it attributable to elevation, which is a *baseline* terrain variable; it changes
 115 mean transfer by $+0.014$, whose interval spans zero. Both figures are post-
 selection, the two predictors having been chosen on the same reversal analysis
 (Section 4.6b). A local cost is measured and no compensating transfer gain is,
 which is weaker than an exchange and is stated as such. The within-region
 half is not itself novel, since comparable results exist for these landscapes
 120 [1, 12]; the paired contrast against portability is. Dimarco et al. [7] transfer
 a predominantly spatially stationary predictor set successfully across a com-
 parable Mediterranean design, and this is the complementary half of that
 picture rather than a contradiction of it.

Contribution 2. The loss is invisible to the diagnostics that

125 **could be run before deployment, and visible to one that cannot.**

Twenty candidate transferability diagnostics from four families are evaluated against the observed transfer outcomes under one bootstrap framework. Area-of-applicability-style predictor-space dissimilarity [21, 22, 16], climatic and geographic distance, learned domain separability, niche-overlap statistics

130 and burn-pattern regime distances all fail to order the matrix; the domain classifier is at ceiling, with $AUC \geq 0.96$ for every pair. Only two diagnostics have intervals excluding zero, and both measure conditional direction agreement, whether each predictor’s signed association with burning points the same way in both regions (Spearman $\rho = +0.84$ and $+0.81$ over sixteen

135 directions from eight region pairs). Both are computed on a data-selected subset of predictors, and their all-nine-feature counterparts span zero, so the result rests on that selection and is reported with it. Equalising the families onto a common twelve directions leaves the ordering unchanged, so it is not an artefact of the marginal rows carrying the smaller sample. Signed asso-

140 ciations need burned labels on both sides, so this is a mechanism diagnosis rather than a pre-deployment screen, and the marginal family, which is the only one that can be run before deployment, is the one that fails. The same point survives without any ranking, at the point estimates: the pair with the highest burned-niche overlap in the matrix fails in both directions while

145 the pair with the lowest transfers in both, so similarity is not sufficient for transfer. That contradicts an expectation the fire literature carries [7, 15] and agrees with what species distribution modelling has found [38, 32]. Shift decomposition in applied remote sensing is not itself new [11]; what is added here is the fire application and the head-to-head of marginal, niche-overlap

150 and conditional diagnostics on the same pairs.

Contribution 3. The mechanism is a sign reversal, and it survives holding geography fixed. Predictors do not merely weaken across regions, they reverse the direction of their association with burning, which is why a distance in predictor space cannot see the failure. The sharpest reversal is elevation, whose association points opposite ways in Manavgat and in Bejís and Muğla with disjoint bootstrap intervals. The same reversal appears inside a single study area, between two fires eleven months apart on an identical grid, where season and population also differ but place does not (Section 4.8); that arm rests on one fire and eleven positive-carrying 5 km blocks, below the floor this design sets for itself, so it corroborates the mechanism rather than establishing it. Two consequences follow and are reported as supporting rather than leading results: label-free alignment by standardisation and covariance alignment [35] does not repair transfer but compresses it towards chance in fourteen of twenty directions, which we believe is the first application of covariance alignment to fire susceptibility; and because the residual gap is conditional, the resource that closes it is target labels, whose price is measurable, thirty-two labelled 5 km blocks recovering 85 to 89 % of the target’s own ceiling in three of six directions tested, 51 to 57 % in two more and 30 % in the sixth.

170 Alongside these, a leakage-audited, spatially blocked evaluation and transfer protocol is released with code, configuration and frozen outputs, so that most of this result can be re-run rather than taken on trust. One component is an exception and is named in the data-and-code availability statement: the reproduction-check code is not in the released repository at the commit

175 of record. The few-shot export, previously also listed as an exception, was
regenerated at commit 6d7a6a71 and its validator now passes on every check.
This matters given evidence that wildfire transfer conclusions are sensitive
to evaluation design and task formulation [42].

A companion paper treats the observational layer beneath this one, where
180 preprocessing decisions taken before any model is fitted are shown to carry
their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study
regions, the data, and the modelling, cross-validation, transfer and adapta-
tion protocol. Section 4 reports results. Section 5 discusses them, including
185 limitations. Section 6 concludes.

2. Related work

2.1. Fire susceptibility mapping, and what it reports

Fire susceptibility mapping has been synthesised recently from both method-
ological and application perspectives [39, 13]. The dominant pattern is sta-
190 ble: geospatial predictors are assembled over a study region, paired with a
historical burned-area record, and fitted with a supervised classifier, most
often a random forest [4], and the resulting surface is published with a cross-
validated AUC in the 0.85 to 0.95 range. That figure is a within-region
estimate. Comparable within-region results already exist for the landscapes
195 studied here [1, 12].

2.2. Pre-fire thermal dryness

Satellite thermal observation gives repeated access to surface state through
fuel moisture content [44]. Pairing land surface temperature with a vegeta-

tion index was established as a live-fuel-moisture estimator for fire-danger
 200 rating by Chuvieco et al. [6], and the Temperature-Vegetation Dryness In-
 dex [33] formalises that feature space into an internally normalised measure,
 which is why it is often expected to travel between regions better than raw
 temperature. That pre-fire thermal state carries information about subse-
 quent fire is established [17, 19, 18]. Gelabert et al. [8] found dead fine fuel
 205 moisture and its anomalies to be the most influential predictor of human-
 caused ignition likelihood across Europe, testing generalisation by pooled
 fitting with per-site evaluation. What has not been tested is whether a clas-
 sifier built on this class survives strict, label-free application to a region it
 never saw.

210 2.3. *Spatial validation and transferability*

Random cross-validation over spatially autocorrelated cells inflates skill
 estimates [31, 29], and spatially blocked designs are the standard remedy
 [37, 23]. The remedy is not uncontested: blocked schemes have been argued
 to introduce pessimistic bias of their own [40, 24, 5]. This paper uses spatial
 215 blocking throughout and reports every result at three block sizes, so that a
 reader can see which verdicts depend on the choice.

Blocking makes a within-region estimate honest but says nothing about
 another region. For that, the field’s instrument is the area of applicabil-
 ity and related predictor-space dissimilarity measures [21, 22, 16], which ask
 220 whether a target’s predictor values fall inside the training data’s envelope.
 Species distribution modelling has examined transferability far more system-
 atically than fire has [43], and two studies there report that geographic and
 environmental similarity do not reliably predict transfer success [38, 32], and

the transferability of satellite-derived empirical models is an active question
225 in fire remote sensing itself [20].

2.4. Dataset shift and label-free adaptation

The canonical taxonomy separates covariate shift, where the predictor
distribution moves but the predictor-response relationship holds, from concept
shift, where the relationship itself changes [25]. Covariate shift is in
230 principle correctable without target labels, by per-region standardisation or
by covariance alignment such as CORAL [35], and domain adaptation has
an established remote-sensing literature [36, 28]. Concept shift is not: no
realignment of input distributions can repair a reversal in the sign of an
association, because detecting it requires the labels being withheld. Shift
235 decomposition in applied remote sensing is not itself new [11]. We are not
aware of a prior application of covariance alignment to fire susceptibility, fire
occurrence or burned-area prediction.

2.5. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [30] pro-
240 vide the long-standing reference point that meteorologically derived danger
indices do not port cleanly between fire environments. WildfireGenome [15]
performs a leave-one-county-out transfer matrix across seven ecologically di-
verse US counties and reports strong within-county performance with highly
variable off-diagonal transfer, ecologically similar pairs transferring well and
245 dissimilar pairs collapsing; its label, however, is a principal-component com-
posite of federal hazard *indicators* rather than observed burned area, and it
applies no adaptation and no shift decomposition. Xu et al. [42] argue that

wildfire transfer conclusions depend strongly on evaluation design and task formulation, a caution that applies here as much as anywhere, and which
250 we address by fixing the analysis protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis. That log is not a formal pre-registration and no independent timestamped registration exists; Section 3.11 states what was fixed and when. For infrastructure at this scale, Kondylatos et al. [14] provide Mesogeos, a 1 km Mediterranean
255 datacube built for data-driven wildfire modelling.

The nearest neighbour, and the contrast the present paper draws. Dimarco et al. [7] is the closest Mediterranean analogue. They harmonise 500 m predictors across four Mediterranean countries, fit tree ensembles under five-fold spatial cross-validation with an 80/20 hold-out, and test transfer
260 both leave-one-country-out and as a full 4×4 matrix. Every transfer exceeds AUC 0.80. Transfers between bioclimatically similar countries score higher, and transfers to Morocco are systematically lower, which they attribute to anthropogenic drivers, fire management practice and data reporting rather than to climate alone. They apply no domain adaptation.¹

265 The two studies share their essential design: Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation and an explicit transfer matrix. Theirs covers four countries and twelve ordered pairs, ours five regions and twenty ordered directions. **What differs is the predictor class.** Their model rests on quantities that are attributes
270 of a place: terrain slope, night-time lights, human modification, population

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

density, and temperature, wind and humidity as long-term seasonal climatologies. Every one is spatially stationary. Ours rests on the state of a surface in one season. Their result and ours are therefore not in conflict; read together they suggest that the relationship between domain similarity
275 and transfer success is **predictor-class dependent**, so that models built on stationary attributes track similarity while models built on dynamic state need not. We present that reading as a live disagreement rather than a settled result: Vesk et al. [38] and Rousseau and Betts [32] align with it from species distribution modelling, while Dimarco et al. and WildfireGenome run
280 against it.

Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology [26]. We follow their Methods, which we take to be authoritative, and note the discrepancy so that the comparison is transparent. Every statement
285 about their work was checked against the full text, read 2026-08-08, not the abstract alone.

3. Data and methods

3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed: Manavgat 2021 and
290 Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular area of interest in EPSG:4326, defined from place coverage rather than from a fire perimeter, and not tuned on burned prevalence, gate outcome or any model metric. One AOI choice was label-informed and is stated as such: the North

295 Evia box was extended after the legacy box was found to carry an atypically high burned prevalence, the extended geometry was then defined from place anchors, and the legacy variant is retained as a sensitivity (Section 4.7a). The AOIs are deliberately not clipped to fire perimeters, so that unburned cells surrounding each fire form the negative class rather than being excluded
 300 by construction. A sixth region, Kozan 2023, is carried as a negative control and is excluded from all modelling by the gate of Section 3.3.

Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Window lengths are matched within
 305 a region and vary between them with the event: 56 to 60 days for predictors and 35 to 49 days for labels. A four-year baseline of window-symmetric composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, window dates and baseline years are registered in `core/regions.py` and are
 310 reproduced in Table 1.

3.2. Burned-area label and the ~ 500 m analysis grid

Labels come from MODIS MCD64A1 Collection 6.1 (`MODIS/061/MCD64A1`) retrieved through Google Earth Engine [10], whose omission and commission characteristics [3] bound every model here. The product is exported onto
 315 the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which duplicates each native ~ 500 m observation across the 30 m pixels beneath it.

The analysis grid is therefore **reconstructed rather than native**: the 30 m grid is partitioned into non-overlapping blocks of `round(500 / 30) =`

320 17 × 17 pixels, giving a nominal cell edge of 510 m, and each cell is identified by its integer block indices `row_500m` and `col_500m`. This approximates the MODIS sinusoidal cell rather than reproducing it, being anchored to the Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a
325 MODIS cell’s 0.250 km²; every block-size label in this paper is therefore the north-south dimension. The companion paper develops what follows from that.

A cell’s representative burn date is the **mode** of the positive sub-pixel day-of-year values in the block, tested against the region’s label window. Be-
330 cause the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. The fraction of positive sub-pixels agreeing with the modal date is recorded as `burn_date_pixel_agreement_fraction` but no agreement threshold is imposed. The label never affects a cell’s eligibility for modelling: unburned, all-no-data
335 and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned inside a region’s own predictor window are removed from its analysis universe rather than counted as unburned, which ran for Muğla, Evia and Montiferru, is recorded as not run for Manavgat and has no recorded status for Bejís.
340 Burning in earlier years is screened for no region in the five-region cohort; the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.14.

3.3. Burned-landcover admissibility gate

345 Before any modelling each region passes a gate that answers one question: of the cells labelled burned, what fraction is dominated by natural vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a cropland-dominated control when the cropland fraction reaches 0.50 instead.

350 The gate uses ESA WorldCover classes aggregated to the same reconstructed cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

3.4. Predictor variables

355 Ten predictors are used, four baseline and six thermal, all summarised per cell over the predictor window. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/061/MOD11A1.

- 360 • **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline’s one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- **Land cover** is ESA WorldCover v200 [45], entering as the dominant class code.
- 365 • **Current LST**, the predictor-window median of Landsat surface temperature in °C.

- **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- 370 • **TVDI and TVDI difference**. The Temperature-Vegetation Dryness Index [33] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain. **TVDI difference** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference`
375 `= current_tvdi - mean(baseline_tvdi)`, in index units rather than standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.
- **Downscaled LST and fused LST**. A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30
385 m grid; the fused product equals observed Landsat LST wherever that is valid and the downscaled surface only where it is not, so gap-filling never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler’s own inputs include coordinates, which is the one route by which a coordinate-derived surface
390 re-enters a feature set from which Section 3.13 excludes coordinates;

the companion paper measures what the increment is worth without these two channels.

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~ 510 m cell, computed from valid
395 30 m pixels only, with the valid fraction recorded. A cell is `valid_for_modeling` when it meets a 30 % valid-pixel floor and no thermal channel is missing. Two populations are then defined. The **primary** population is natural vegetation, cells whose combined tree, shrub and grass fraction reaches 0.50, which excludes cropland from every burnable mask. The **sec-**
400 **ondary** population is all valid cells, reported throughout as a sensitivity. Every headline result is given for both.

3.6. Feature sets and classifier

Two nested feature sets are compared. The **baseline** set is NDVI, elevation, slope and land cover. The **thermal** set is the baseline plus the
405 six thermal channels. Land cover is one-hot encoded and numeric features are standardised inside the fitting pipeline. The classifier is a random forest [4] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison in this paper
410 is confounded by a model choice.

3.7. Spatial-block cross-validation and bootstrap uncertainty

Cross-validation is spatially blocked. Blocks are squares of the analysis grid formed as `row_500m // B` by `col_500m // B`, and folds are drawn over

whole blocks, so no block is split between training and test. Five folds are
 415 used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10
 km, for the within-region analysis.

Uncertainty is a spatial-block bootstrap: blocks are resampled with re-
 placement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the
 interval. Because it is blocks that are resampled, what bounds an interval’s
 420 reliability is the number of blocks carrying at least one burned cell, and those
 counts are reported alongside the intervals. Where a verdict rests on too few
 such blocks it is stated as indicative rather than as an interval.

3.8. Cross-region transfer protocol

For each ordered pair of regions the model is fitted on the source pop-
 425 ulation and applied to the target population with **no target label used
 at any point**: no refitting, no threshold selection, no calibration. Both
 the baseline and the thermal feature set are transferred, so that the thermal
 block’s contribution is a paired difference within a direction rather than a
 comparison across directions. All twenty ordered directions among the five
 430 regions are computed.

3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.

Region-wise z-score. Each region’s numeric features are standardised
 using its own statistics, source statistics from source data and target statistics
 435 from target data, never pooled. The classifier is refitted on the z-scored
 source and applied to the z-scored target. This removes first- and second-
 order marginal offsets.

CORAL after region-wise z-score. Starting from the z-scored features, the source covariance is aligned to the target covariance by the standard whitening-recolouring map [35], with regularisation $\lambda = 10^{-5}$ and eigenvalues floored at 10^{-12} . Critically **the transform is applied to the source only**; the target is left exactly as it is. The classifier is refitted on the aligned source. Neither variant sees a target label, and both are verified label-blind at run time.

λ sensitivity was assessed over nine values from 0 to 10^{-1} on four directions in both feature families, moving transfer AUC by at most 0.014 in any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies outside the released sweep; the companion paper reports it.

3.10. Transfer-gap decomposition and the concept-shift diagnostic

For each direction the gap between the target’s own within-region skill and the raw transfer result is split into a part the best label-free adaptation recovers and a part it does not, with the recovered fraction defined as $(\text{adapted} - \text{raw}) / (\text{within} - \text{raw})$, signed and unclipped, and its interval from the same paired bootstrap. The recovered part bounds what covariate-level correction can achieve; the remainder is the conditional residual.

The mechanism is diagnosed by **signed univariate association**. For each numeric predictor the raw ROC-AUC of that predictor against **burned** is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a value below 0.5 is read as a direction rather than as weakness. A reversal is called bootstrap-supported only when the two regions’ intervals are disjoint, under the same 10-cell spatial-block bootstrap.

3.11. *Transferability diagnostics versus transfer*

Twenty candidate diagnostics from four families are each rank-correlated against the same target quantity, the raw thermal transfer AUC over the
465 twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

The **marginal** family includes area-of-applicability-style dissimilarity in
470 scaled, importance-weighted predictor space [21, 22, 16], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [34] and Warren’s I [41] over the burned cells of each region. The **regime** family includes burned-patch component counts and effective component counts. The
475 **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

Two properties of this design are stated in advance because they bound
480 what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are evaluated at that power with no family-wise error control claimed; where a
485 diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

3.12. Interventions

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and
490 evaluated on the held-out region, with folds blocked as before. This asks whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors whose signed association reverses between regions with bootstrap support are **elevation_mean** (Manavgat against Bejís and against Muğla) and **lst_anomaly_mean**
505 (Bejís against Evia). They are dropped and everything is refitted, within-region and across every direction, and the two are also dropped singly so the cost can be attributed. This measures what the removal costs locally and what it returns on transfer on the same footing. Note that the two features are selected by the same reversal analysis against which the result is then
500 read, so both quantities are post-selection estimates and no correction for that selection is applied.

3.13. Leakage control

An explicit forbidden-column set is enforced at every model fit. Coordinates (**lon**, **lat**, **row**, **col** and their normalised forms), every burn-date and
505 label-provenance column, and the agreement fraction are excluded from all feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

Reproducibility. All randomness uses seed 42 and the bootstrap uses
510 1000 replicates throughout. The transfer and adaptation analysis runs in an

environment separate from the upstream pipeline’s, so every within-region model was refitted there and compared against the frozen upstream output, and the independently implemented adaptation against the pipeline’s own; the within-region comparisons agree exactly and the twenty transfer directions to within 1.6×10^{-7} . Because random-forest fits are not bit-identical across library versions, all numbers here were produced under, or verified against, scikit-learn 1.9.0, and cross-region point estimates carry an implementation tolerance of about ± 0.02 to 0.03 if that version is not fixed. The companion paper reports the version sensitivity and the reproduction check in full.

Sensitivity analyses. Every headline result is repeated across two analysis populations, three spatial-block sizes, the CORAL regularisation sweep and both feature sets. Where a conclusion depends on one of those choices, the dependence is reported rather than resolved by choosing the favourable setting.

3.14. *Same-geography event-to-event comparison (Muğla 2021 versus 2022)*

Muğla is the one region for which a second fire event is analysed on the identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented as one. The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no

difference can be attributed to the year alone. The registered claim is same-
geography event-to-event. The pair is deliberately kept out of the twenty-
direction matrix, and that exclusion is enforced in released code rather than
540 merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion
applies to the 2022 arm alone, masking every cell that burned in 2021 so the
previous year’s scar does not enter the following year’s analysis: it removes
3,073 cells, of which 2,941 belong to the primary population, taking it from
545 41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of
38,790 cells and three byte-identical static predictors, and in the 2022-to-2021
direction every target positive lies outside the source training population.
Section 4.8 states what follows for how these two numbers may be read.

4. Results

550 4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate as wildfire can-
didates, and the negative control fails it as intended. Kozan 2023 returns a
natural-vegetation fraction of 0.017 against the 0.50 threshold and the verdict
cropland-dominated control, and is excluded from all modelling. The sepa-
555 ration is not marginal: the five admitted regions carry natural-vegetation
fractions of 0.723 to 0.991, so the threshold falls in an empty interval rather
than between neighbouring cases. This establishes that the gate discrimi-
nates burned area produced by natural-fuel combustion from burned area
produced by post-harvest stubble burning, which MCD64A1 does not distin-
560 guish.

Table 1: Region summary (recap of Table 1; final numbering at assembly). Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, **burnable_tree_shrub_grass and valid_for_modeling == True**, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Gate verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejis 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monticelli 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

North Evia is analysed on an extended AOI. Relative to the legacy box, the extended box leaves the burned scar essentially unchanged while cutting overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to 0.287. The effect of that choice on transfer is reported in Section 4.7a.

565 *4.2. Within-region: the thermal increment replicates in five regions*

In every region, adding the six thermal predictors to the baseline increases spatially blocked out-of-fold ROC-AUC. The increment’s bootstrap interval excludes zero in all five regions at 1 km and at 5 km blocking, which are the two scales this design supports as intervals. At 10 km the point estimates
570 hold, from +0.048 to +0.154, but they rest on 6 to 33 positive-carrying blocks and are reported as indicative rather than as intervals, for the reason given in the table note.

Table note (resampling units). The bootstrap resamples spatial blocks, so what bounds an interval’s reliability is the number of blocks that carry
575 at least one burned cell. Those counts fall quickly as blocks coarsen. At 2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10 cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At 20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No bootstrap
580 replicate was invalid at any block size except Bejís at 20 cells, where 5 of 1000 were single-class — which is the symptom of the six positive-carrying blocks just reported, and a further reason to read that row as indicative. An equal-tailed percentile interval built on six positive-carrying blocks has no meaningful coverage, and Montiferru at 20 cells additionally feeds only 12
585 groups into a 5-fold grouped split, so its models train on about ten blocks

Table 2: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.8); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

each. **The 20-cell row of this table should be read as indicative rather than as an interval.** The 10-cell row, where every region has 16 to 70 positive-carrying blocks, is the coarsest blocking this design supports properly, and the increment holds there in all five regions. Source: `paper/`
590 `referee2_numbers.md`, block C, counted from the frozen per-cell prediction tables.

This is the first half of the trade-off. It is not itself novel, and it is reported because the second half is measured against it.

4.3. Cross-region transfer, and what label-free adaptation does to it

595 **Raw transfer is heterogeneous and includes anti-predictive directions.** Raw target AUC spans 0.326 to 0.686. Twelve of 20 directions are above chance with CI support, six are *below* chance with CI support (both directions of $\text{Manavgat} \leftrightarrow \text{Bejís}$ and $\text{Manavgat} \leftrightarrow \text{Muğla}$, plus $\text{Bejís} \rightarrow \text{Evia}$ and $\text{Evia} \rightarrow \text{Bejís}$), and two intervals span 0.5. Even the best raw transfer ($\text{Evia} \rightarrow \text{Manavgat}$,
600 0.686) remains far below the target’s own within-region thermal performance (0.870); across all directions the raw deficit against the within-region reference is 0.184 to 0.592 AUC. Those counts belong to the 2-cell blocking of Table 4. At the more conservative 10-cell (~ 5 km) blocking the same points give 9 above, 4 below and 7 uncertain, with no direction changing side of
605 the chance line (Section 4.7g). Four of the six below-chance directions keep their support there: $\text{Manavgat} \rightarrow \text{Bejís}$, $\text{Muğla} \rightarrow \text{Manavgat}$, $\text{Bejís} \rightarrow \text{Evia}$ and $\text{Evia} \rightarrow \text{Bejís}$. $\text{Bejís} \rightarrow \text{Manavgat}$ and $\text{Manavgat} \rightarrow \text{Muğla}$ lose it and carry no verdict. The qualitative statement is unchanged. The counts should not be read as exact.

610 **The static baseline does not transfer either.** This is the control

for the interpretation the rest of the paper invites, and it constrains that interpretation sharply. Running the same twenty directions with the terrain, fuel and greenness baseline alone gives a mean target AUC of **0.537**, against **0.541** for the thermal model. The predictor class that this paper’s framing treats as the portable one — static attributes of a place, the class Dimarco et al. transfer successfully — is itself barely above chance here. The paired per-direction contrast is reported in Section 4.6b; what matters at this point is that the transfer failure documented below is not specific to the dynamic block. Adding pre-fire thermal state to a baseline that does not travel produces a model that does not travel.

The thermal block’s paired contribution to transfer, with its interval. Differencing the two matrices direction by direction gives a mean of **+0.004**. The directions are not independent — each region appears in eight of the twenty — so the interval depends on what is treated as the resampling unit, and all four units the design permits give the same answer:

Table 3:		
Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive	20	$[-0.027, +0.034]$
Unordered pairs, cluster bootstrap	10	$[-0.028, +0.036]$
Unordered pairs, t on pair means	10	$[-0.034, +0.042]$
Regions, leave-one-out jackknife	5	$[-0.037, +0.046]$

Every interval spans zero and the point estimate is a small fraction of each width. The per-region jackknife also shows how little the mean is anchored: holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn

gives +0.0148, +0.0021, +0.0065, **-0.0081** and +0.0060, so **dropping Evia**
630 **alone reverses the sign of the headline.** Within-direction sampling variability is not propagated into any of these four; they resample between directions only.

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty directions
635 span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the raw spread, with no adapted direction exceeding 0.631 against within-region references of 0.859 to 0.918. Adaptation raises the failing directions and degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began
640 and 6 end further from it; five of those six involve Montiferru, the smallest and last-added region, and move upward, while the sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (x) in Section 5.9, since Bejís→Manavgat is counted as compressed on a margin of 0.001.

645 In the six directions where raw transfer was below chance, the best label-free method recovers at most 34 % of the gap to the within-region reference, so the remaining conditional fraction is at least 0.66 everywhere. Seven directions show *negative* recovery, meaning adaptation moves the score away from the reference; in six of those raw transfer was already above chance and
650 adaptation destroyed that advantage. Label-free alignment therefore does not act as a repair mechanism.

Table 4: Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

Table 5: Transfer-gap decomposition (four-AOI set, 12 directions). Within = target’s within-region thermal AUC; best adapted = the better of z-score/CORAL; recovered fraction = (adapted – raw)/(within – raw), signed and unclipped, with paired bootstrap CI (1000 replicates). Montiferru directions are not part of this decomposition (per-pair absolute decompositions exist without fraction CIs). The status column asks whether the *adapted* value clears chance and uses the 2-cell adapted intervals of Table 4. The adapted arms were not recomputed at the coarser blocking of Section 4.7g, which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.550 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

4.4. Transferability diagnostics: only conditional similarity orders transfer

Twenty candidate diagnostics from four families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty
655 ordered directions, under one common pair-based bootstrap.

Table note (power): the effective sample is 10 unordered pairs (6 for the 12-direction rows; 8 for the supported-conditional rows). The two directions of a pair are not independent and every pair shares regions with three others. Intervals of width ± 0.5 to 0.8 cannot rule out moderate true correlations;
660 *null rows are "not shown to order transfer", not "shown not to".*

Only two diagnostics have intervals excluding zero, and both are conditional. The stronger is the sign-agreement fraction over interval-supported features, at $\rho = +0.84$ [$+0.58, +0.88$]; the cosine variant reaches $+0.81$. Every marginal measure fails, including area-of-applicability-style
665 dissimilarity in predictor space, climatic distance and geographic distance, and so do the niche-overlap and regime families. The learned domain classifier is at ceiling, separating source from target at $\text{AUC} \geq 0.96$ for every pair, which makes it useless as an ordering instrument precisely because it always succeeds.

670 Two limits are stated with the result rather than after it. The conditional index is evaluated at its own design size: its tie structure caps the achievable Spearman at $+0.861$, the observed $+0.840$ therefore sits essentially on that ceiling, and the exact one-sided permutation p is 0.0060 , the smallest this tie structure can produce, against a Bonferroni threshold of 0.0026 over the
675 nineteen computed variants. No outcome of this diagnostic could have cleared family-wise correction on ten effective pairs. And signed associations require

Table 6: All transferability diagnostics versus raw thermal transfer (20 ordered directions). Spearman ρ with pair-based bootstrap 95% CIs. Exp. = expected sign. Rows with n = 12 exist only for the four-AOI subset. The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y x) conditional	+	16	+0.84 [+0.58, +0.88]	yes
Cosine, supported features	P(y x) conditional	+	16	+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	P(y x) conditional	+	20	+0.50 [−0.17, +0.83]	no
Vector	P(y x) conditional	+	20	+0.27 [−0.36, +0.77]	no
Spearman, all 9	conditional				
Agreement count, all 9	P(y x) conditional	+	20	+0.18 [−0.40, +0.72]	no
Schoener’s D, 1-D mean	P(x y=1) niche	+	20	+0.24 [−0.45, +0.74]	no
Warren’s I, 1-D mean	P(x y=1) niche	+	20	+0.22 [−0.42, +0.73]	no
Schoener’s D, PCA-2D	P(x y=1) niche	+	20	+0.10 [−0.51, +0.68]	no
Warren’s I, PCA-2D	P(x y=1) niche	+	20	−0.07 [−0.66, +0.49]	no
Mahalanobis, burned centroids	P(x y=1) niche	−	20	−0.23 [−0.75, +0.44]	no
Domain-classifier AUC	P(x) marginal	−	20	−0.32 [−0.78, +0.33]	no
Predictor-space mean dissimilarity	P(x) marginal	−	35		
		−	12	−0.10 [−0.54, +0.43]	no
Predictor-space p95 dissimilarity	P(x) marginal	−	12	−0.08 [−0.59, +0.49]	no

burned labels in both regions, so the family that works is not available before deployment while the family that fails is.

A third limit is a selection rather than a sample-size issue, and it is the sharpest of the three. The two rows that clear zero are the *supported-feature* variants, where the feature subset is chosen by whether two regions' bootstrap intervals happen to be disjoint — a data-dependent selection made on the same data, with no correction. Their unselected counterparts over all nine features are $\rho = +0.50$ $[-0.17, +0.83]$ for the cosine and $\rho = +0.18$ $[-0.40, +0.72]$ for the agreement count, both spanning zero. The result lives in the selection step, and is reported as such.

The families are compared on unequal samples, and equalising them does not change the ordering. The marginal, applicability, climatic and geographic rows sit on twelve directions, because those diagnostics exist only for the four-region subset; the supported-conditional rows sit on sixteen and the rest on twenty. A reader may reasonably ask whether "the marginal family fails" is a statement about power rather than about diagnostics. Recomputing every row on the common twelve directions answers it: the published values reproduce to 4.8×10^{-5} , the conditional rows still lead at $+0.87$ $[+0.65, +0.88]$ and $+0.85$ $[+0.43, +0.88]$, and every marginal row still spans zero. The ordering is not an artefact of unequal samples.

4.5. The contrast pair: similarity is not sufficient

The clearest single view of Table 6 needs no ranking at all. Manavgat and Muğla lie in the same country and fire year, 307 km apart by the centroid geodesic distance this paper uses as a diagnostic (their nearest boundaries are 191 km apart), and their burned cells occupy the most similar environmental

envelope of any pair in the matrix, with per-feature Schoener’s D of 0.77 to 0.89. Yet five of nine feature-response directions point opposite ways, including elevation, whose signed AUC is 0.374 [0.289, 0.471] in Manavgat
705 against 0.611 [0.532, 0.690] in Muğla with disjoint intervals, and all four absolute thermal channels. Transfer is below chance in both directions at the point estimate, 0.470 and 0.401. Both directions also sit deep inside the nominal area of applicability, at 0.875 and 0.531 of target cells inside the weighted region.

710 Bejís and Montiferru sit at the opposite extreme, with burned envelopes that barely overlap and the most dissimilar values on every overlap measure, and they transfer above chance in both directions. That half of the contrast rests on point estimates, since neither direction carries a verdict at 5 km blocking, and the marginal applicability audit was never produced for
715 Montiferru. The claim is one of *sufficiency*: similarity does not guarantee transfer, and dissimilarity does not preclude it. Established by coexisting counterexamples, it does not depend on the number of pairs available.

4.6. Interventions: pooling and feature removal obey the same conservation

(a) **Pooled multi-region training.** Training on the pooled populations
720 of four regions and testing on the held-out fifth does not rescue transfer. At the point estimate the pooled thermal model never beats the best single-source pairwise transfer for any target, with shortfalls of 0.02 to 0.22, and it remains 0.28 to 0.50 AUC below the within-region ceiling. For two targets it falls below the pairwise mean and below chance, though only Bejís is below
725 chance with interval support, at 0.417 [0.369, 0.467].

(b) **Removing the direction-reversing features.** The two predictors

whose signed association reverses between regions are **elevation_mean** and **lst_anomaly_mean**. Retraining without them costs -0.081 of mean within-region AUC, supported in every region (per-region deltas -0.060 , -0.130 , -0.074 , -0.063 , -0.079 , every interval entirely below zero), and changes mean transfer by $+0.014$ $[-0.017, +0.045]$, which spans zero.

Two qualifications belong with those numbers rather than after them. First, the debit is not the thermal block’s. Dropping elevation alone accounts for -0.061 of it ($0.888 \rightarrow 0.827$) and dropping the LST anomaly alone for -0.013 ($0.888 \rightarrow 0.875$), so roughly three quarters of the cost is the removal of a **baseline** terrain variable, not of a thermal channel. Second, both figures are post-selection: the two predictors were chosen because they reverse, using the same data on which the -0.081 and the $+0.014$ are then estimated, and no correction for that selection is applied.

Table 7:

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop elevation_mean	0.827	0.546
drop lst_anomaly_mean	0.875	0.544
drop both	0.807	0.556

What this measures, stated at the strength it supports, is a local cost with no compensating transfer gain. It does not measure an exchange, because the transfer side is a null on both arms: the thermal block’s own contribution to transfer is $+0.004$ with an interval spanning zero (Section 4.3), and removing the reversing predictors returns $+0.014$ with an interval spanning zero. Two nulls on the portability axis do not constitute a price paid.

4.7. Sensitivity analyses

(a) **Evia AOI and prevalence.** Repeating the raw transfer arms with the legacy, high-prevalence Evia box leaves every qualitative conclusion unchanged: thermal raw transfer AUCs move by at most a up to 0.07 — the
750 largest is Evia→Bejís, 0.378 to 0.448 — and no direction changes side of the chance line.

(d) **CORAL regularisation.** Over nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family, so no CORAL-dependent con-
755 clusion here is sensitive to λ in that range. The companion paper reports the canonical $\lambda = 1$, which the released sweep omits.

(g) **Blocking scale.** Recomputing the transfer quantities at 10-cell ($\approx$ 5 km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncer-
760 tain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes support from six verdicts and adds none.

The point estimates are unchanged, but that is an identity rather than a result and should not be offered as robustness: the blocking scale is the bootstrap *resampling unit*, and each point estimate is computed once over
765 all target cells, so no choice of block size could have moved one. What the comparison does establish is the direction of the fragility — every verdict that changes, changes towards "no verdict" — and that no direction crosses the chance line under the widened intervals. The counts are the fragile part of this paper. The sign pattern is not thereby shown to be robust; it is simply
770 not tested by this particular variation.

4.8. *The same geography, a second fire: reversal with place held constant*

Every result above compares different places. Muğla admits a stricter test, because a second fire occurred inside the identical AOI on the identical analysis grid: the 2021 event, with its predictor window closing on 28 July, and the 2022 event, whose matched window closes on 20 June. Region, bounding box, cell definition, feature registry and processing chain are the same. What the design holds fixed is place; what it does not hold fixed is season, since the 2022 fire ignites about six weeks earlier, nor population, since the 2022 arm is defined by removing the 2021 scar.

Elevation reverses with bootstrap support. In 2021 higher ground burned preferentially, at 0.611 with its interval entirely above 0.5; in 2022 lower ground did, at 0.296 with its interval entirely below. The two intervals are disjoint, 0.532 against 0.355, and the difference is -0.317 $[-0.414, -0.220]$. The same predictor, the same region, the same grid, and an association that points the opposite way in two fires eleven months apart. The four absolute thermal channels change side as well, from bootstrap-supported lower-values-burn in 2021 to higher-values-burn point estimates in 2022.

Two structural properties must be read alongside these numbers, because they have no analogue in the twenty-direction matrix. The two arms are not disjoint samples: they share 38,789 of the 2022 arm's 38,790 cells, and elevation, slope and land cover are identical to the digit across all 73,098 grid cells, so only NDVI and the six thermal channels carry new information between them. And the removal is the target's own positive class, so in the 2022 to 2021 direction not one target positive is present in the source training population while 38,789 of 38,819 target negatives are, and membership of

Table 8: Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against **burned**, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.14). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 2’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 2 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

the source training set alone separates the 2021 target’s classes at ROC-AUC 0.9996. We therefore cannot bound what that asymmetry does to the two transfer numbers, only state that a direction with this structure is not comparable to the twenty between-region directions. The elevation reversal is the finding; the structural asymmetry is a competing explanation this design cannot exclude. The known part of the bias runs the safe way: the removed cells are high, with a median of 563 m, and unburned in 2022, so removing them raises the 2022 elevation AUC and makes the reversal smaller rather than larger.

4.9. *What target labels cost: the recovery curve*

Everything above measures a failure; this prices it. If the residual gap is conditional and label-free alignment cannot close it, the missing resource is target-conditional information, and the direct way to supply it is target labels. The frozen few-shot diagnostic answers how many for three regions in all six ordered directions, using one 10-cell (≈ 5 km) spatial block as the unit of labelling effort, budgets of 0 to 32 blocks drawn under a fixed tier order with seeds independent of any result, and ten repeats per budget. Recovery is read against a matched target-only ceiling of 0.777 to 0.824 under the same folds and blocking.

Thirty-two labelled blocks recover **85 to 89 % of the target’s own ceiling in three of the six directions**, two of which started below chance and the third at 0.583; 51 to 57 % in two more, which are the directions where the conditional gap is widest; and 30 % in the sixth. That budget is 7 to 20 % of the target’s natural-vegetation population, so it is a real answer to "what do we do" and not a cheap one. At small budgets the same

intervention damages the direction that already transfers best without any labels. Six directions in three regions cannot support a general label budget and none is offered.

5. Discussion

825 5.1. *Principal findings*

Three results carry this paper. First, the pre-fire thermal block raises spatially blocked within-region ROC-AUC by +0.056 to +0.153 in every one of five Mediterranean regions, with bootstrap support at 1 km and 5 km blocking, and contributes +0.004 $[-0.028, +0.036]$ across twenty ordered
830 transfer directions, indistinguishable from zero and with a sign that varies by pair. The static baseline transfers no better, at a mean of 0.537 against 0.541, so the failure is a property of this matrix rather than of the dynamic block specifically. Removing the two reversing predictors, elevation and the LST anomaly, costs -0.081 of within-region skill, supported in every re-
835 gion and three quarters attributable to elevation, and changes transfer by +0.014 $[-0.017, +0.045]$, which also spans zero. Second, of twenty candidate transferability diagnostics only two have intervals excluding zero and both are conditional, while every marginal measure fails, including the area-of-applicability family that is the only one runnable before deployment. Third,
840 the mechanism is a reversal in the sign of the predictor-burning association, and it persists inside one study area across two fires.

5.2. *Why the thermal increment is real but local*

The within-region increment is not an artefact to be explained away. It replicates in five independent regions, survives coarsening of the spatial

845 blocks to ~ 5 km with its interval intact, persists in the secondary all-valid population, and survives a predictor window closed 7 and 14 days earlier in all five regions. The direction of that last result is region-specific and should not be overstated: the contribution strengthens in Bejís ($0.058 \rightarrow 0.079$ at 14 days) and Muğla ($0.115 \rightarrow 0.128$), is flat in Montiferru, and **weakens**
850 **monotonically in Evia** ($0.156 \rightarrow 0.149 \rightarrow 0.135$). What holds everywhere is survival, not improvement. The transfer failure is therefore not evidence that the thermal signal is spurious. It is evidence that the fitted relationship is *local*.

The natural objection is that different Mediterranean regions are simply
855 different systems, with distinct fuels, terrain and fire histories, so that a predictor meaning one thing in one place and another elsewhere is not instability but a comparison of two different systems. The two Muğla events speak directly to that. Region, AOI, analysis grid, feature registry and processing chain are identical, and the static predictors are identical cell by
860 cell, yet elevation’s association with burning reverses with disjoint intervals, $0.611 [0.532, 0.690]$ in 2021 against $0.296 [0.230, 0.355]$ in 2022, a difference of $-0.317 [-0.414, -0.220]$. Holding geography fixed does not stabilise the direction of the relationship.

The mechanism there is physical rather than statistical. The 2021 season
865 burned as a dispersed complex across the region’s full relief; the 2022 event was a single compact scar confined to lower ground. Two caveats attach and neither is dismissed. Season and year are confounded, because the 2022 event ignites about six weeks earlier, and **that confound cannot be resolved in this study area**: the two events sit 42 days apart in median burn day-

870 of-year, neither year contains a second event at the other’s phase, and a
calendar-matched arm would carry nine burned cells against a gate minimum
of thirty. Separating year from seasonal phase needs a region with two events
at a matching phase in different years, and no region here has one. The design
also does not hold the population fixed, since the 2022 arm is the 2021 arm
875 with the 2021 scar removed, which Section 4.8 states in full. What survives
both caveats is the narrower reading that *something* reverses the elevation-
burning association with geography, grid and pipeline held fixed.

The sharpest supported reversal belongs to elevation, a static predictor,
so instability is a property of the predictor-to-burning mapping generally
880 rather than of thermal channels specifically. The thermal block is where the
trade-off is *costly*, not where instability is worst.

5.3. *Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks whether the target’s
predictor values look like the training data’s. That question is orthogonal to
885 the one that matters when the failure is conditional. A target region can sit
deep inside the training envelope while the relationship between those pre-
dictors and burning points the other way, and Manavgat to Muğla is exactly
that case: 0.875 of target cells inside the weighted area of applicability, and
transfer below chance.

890 The same holds for the niche-overlap and regime families. At the point
estimates, the pair with the highest burned-niche overlap in the matrix
fails in both directions while the pair with the lowest transfers in both, so
no monotone function of overlap can order the outcomes. The qualifier is
not decorative: at the 5 km blocking this paper otherwise defends, neither

895 Bejís↔Montiferru direction carries a verdict and Manavgat→Muğla loses its
below-chance support, so this contrast is a statement about point estimates
and is made as one. The domain classifier is at ceiling for every pair, which
is informative in itself: the regions are trivially separable in predictor space
whether or not their transfer works, so separability carries no ordering infor-
900 mation here.

5.4. *What the conditional diagnostic is, and what it is not*

The sign-agreement index reaches $\rho = +0.84$ [+0.58, +0.88] against ob-
served transfer, and it is reported with three limits attached rather than after
the fact. Its tie structure caps the achievable Spearman at +0.861, so the
905 observed value sits essentially on its own ceiling and the ranking it supports
is coarse. Its exact one-sided permutation p is 0.0060, the smallest that tie
structure can produce, against a Bonferroni threshold of 0.0026 over nineteen
computed variants, so no outcome could have cleared family-wise correction
on ten effective pairs. And it needs burned labels in both regions, which
910 makes it a mechanism diagnosis rather than a pre-deployment screen.

That last limit is the substantive one, and it is why the paper's practical
conclusion is not "use this index instead". It is that the family of diagnostics
that can be run before deployment is the family that fails here, and the
information that would order transfer is the information a practitioner does
915 not have when the decision is made. The honest statement is that these
twenty candidates were **not shown to order transfer**, not that they are
shown incapable of it.

5.5. *Interventions obey a conservation pattern*

Both interventions show the same shape. Pooling four regions never beats
920 the best single-source transfer for any target and stays 0.28 to 0.50 below
the within-region ceiling, so aggregation does not manufacture the missing
conditional information. Removing the reversing predictors costs -0.081 of
within-region skill with interval support in every region, and changes mean
transfer by $+0.014$ $[-0.017, +0.045]$, which is not distinguishable from zero.

925 We should be exact about what that pair of numbers is, because it is
easy to read as an exchange and it is not one. The thermal block's own
contribution to transfer is $+0.004$ with an interval spanning zero; removing
the reversing predictors returns $+0.014$ with an interval spanning zero. Both
arms are null on the portability axis. So what the interventions measure is a
930 local cost and no compensating transfer gain — not a conservation law, and
not a rate at which local skill can be sold for portability. No such rate is
estimated here, and the earlier framing of the increment as something that
is "spent" on portability overstated what two nulls can support.

5.6. *The regime hypothesis, reported as it happened*

935 A regime-structure explanation was stated in advance and the data con-
firmed the null. The regime-distance correlation has the wrong sign at the
point estimate, the most regime-similar pair fails in both directions, and the
most regime-different pair transfers above chance. One mundane explanation
can be set aside: Muğla has by far the largest population, but cutting it to
940 Manavgat's cell count, and separately to Manavgat's positive count as well,
leaves its transfer behaviour inside the subsampling range in both roles. The

error was in the hypothesised grouping, not in the data, and with ten pairs this cannot refute regime typology [2] in general.

5.7. *The empirical contrast with Dimarco et al.*

945 Dimarco et al. [7] transfer successfully across a comparable Mediterranean design and we do not, and the two results are not in conflict. Their predictors are attributes of a place, ours describe the state of a surface in one season. Read together they suggest that the relationship between domain similarity and transfer success is predictor-class dependent.

950 That reading is one of at least two, and the alternative should be stated because we cannot separate them here. Their response variable is human-driven **ignition**; ours is burned **area**. Ignition likelihood is dominated by human access and activity, which are stable and near-universal drivers, whereas burned extent is dominated by spread — fire weather, fuel continuity, terrain. A predictor-class explanation and a response-variable explanation for the divergence are therefore confounded in this comparison. Our own data speak against a simple predictor-class reading in any case: the static baseline transfers at a mean of 0.537 here (Section 4.3), so within this cohort the place-attribute class does not travel either. We present that as a live
960 disagreement, since species distribution modelling aligns with it and the fire literature does not, and it is a reading our data support without proving.

5.8. *Implications*

For practice, transfer skill has to be *measured* rather than inferred from similarity, and the pre-deployment diagnostics currently used for that inference
965 did not order it here. Where a model must be moved, the resource that

closes the gap is target labels: thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s own ceiling in three of six directions, and 30 to 57 % in the rest. That is a real answer and not a cheap one, being 7 to 20 % of the target’s natural-vegetation population, and at small budgets the same
970 intervention damages the direction that already transfers best.

For method development, the results point away from better unsupervised alignment. Two label-free methods applied carefully moved fourteen of twenty directions towards chance, and every direction they improved involves the smallest region. A sign reversal is not a distribution mismatch, and no
975 realignment of inputs can repair it.

5.9. Limitations

The transfer failure is a finding, not a limitation. The limitations are the boundaries on how far it generalises.

(i) **No meteorological covariates** enter the models, so we cannot say
980 how the trade-off behaves for a mixed thermal-plus-weather predictor set.

(ii) **Temporal transfer is measured for one region only**, and even there year and seasonal phase are confounded, a confound that cannot be resolved in this study area for the reason given in Section 5.2. Its 331 burned cells also leave the thermal reversals unresolved at interval level, and the
985 pair holds place fixed but not population. Those two arms are additionally the only transfer directions here computed by us rather than read from the pipeline author’s frozen export, with his unmodified code and the same pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1
990 [9], whose omission and commission characteristics [3] bound every model

evaluated here. No second burned-area product covers 2021 and 2022 at this resolution; the companion paper reports what an independent active-fire observation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even
995 after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

1000 (vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of**
1005 **ten region pairs.** Both the successes and the failures of Section 4.4 should be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained.**
It is where the conditional diagnosis bites hardest and where feature removal recovers most. Two candidates have now been tested and neither survives.
1010 Its predictor window was not meteorologically extreme. And the quality-screening difference in its coarse thermal input, whose induced change correlates with elevation at +0.615 and was therefore the more worrying of the two, does not propagate: rebuilding the region’s entire downstream chain from a quality-screened MODIS input changes 22,304 of 24,150 cells’ down-
1015 scaled surface by up to 10.9 °C and moves no signed univariate association by

more than +0.0003, leaving elevation's at 0.374 [0.290, 0.472] in both arms and the population unchanged. That is structural rather than fortunate: elevation is a DEM variable the screening cannot touch, and fusion falls back on the MODIS-derived surface across only 2.14 points of coverage. With
1020 one fire season per region the remaining candidates are not separable in this design.

(ix) **The interval-support counts are less stable than the point estimates behind them.** Several verdicts sit within a thousandth of their reference value, and at 1 km blocking the published split of ten positive,
1025 seven negative and three uncertain turns on a lower bound of -0.00045 . The point estimates and the sign pattern are stable; the counts are not. Every sentence in this paper that leans on an exact count of supported directions should be read at that precision.

6. Conclusions

1030 Pre-fire thermal dryness adds a real and repeatable increment to burned-area discrimination. Across five Mediterranean regions, ROC-AUC was raised by +0.06 to +0.15 over a static and near-static terrain, fuel and greenness baseline. That baseline's only time-varying member is the vegetation-index composite. The gain survived spatial blocking at about 5 km, the coarsest
1035 scale this design supports as an interval. It also survived a predictor window closed up to two weeks before the first labelled burning.

That skill is local. Paired per direction, the same predictor block adds nothing distinguishable from zero to cross-region transfer. The mean is +0.004 over twenty directions and its interval spans zero under all four

1040 between-direction resampling units we computed. Its contribution changes
sign from one direction to another. The paired deltas run from -0.148 to
 $+0.132$, twelve positive and eight negative. At the conservative 5 km block-
ing, five to six directions are helped and three to four are harmed with interval
support, with the rest carrying no verdict. The block is also the swing factor
1045 at the chance line.

The control that fixes the meaning of all this is the baseline arm, which
transfers at a mean of 0.537 against the thermal model’s 0.541. The static
predictor class is not portable here either, so the honest summary is that
nothing in this feature space travels, not that dynamic state is uniquely frag-
1050 ile. Removing the two reversing predictors, elevation and the LST anomaly,
costs -0.081 of mean within-region skill, with interval support in every re-
gion and roughly three quarters of it attributable to elevation, a baseline
terrain variable; it changes mean transfer by $+0.014$, an estimate whose in-
terval spans zero. A local cost is measured. No compensating transfer gain
1055 is, on either arm, so no exchange between the two is demonstrated.

The failure is conditional. The direction of the link between dryness
and burning changes from one region to another. None of the similarity
diagnostics tested here ordered the transfer matrix: predictor-space distance,
domain separability, niche overlap and regime structure all failed. On ten
1060 effective pairs those null results mean not shown to order transfer, rather
than shown not to. It is also why label-blind adaptation compresses transfer
towards chance instead of repairing it, and why pooled multi-region training
does not escape it.

The practical implication is a change in what is checked before a dynamic-

1065 state fire model is transferred. The question is not whether the target region
lies inside the source’s environmental envelope. In our matrix, the pair with
the highest envelope overlap failed in both directions, and the pair with the
lowest overlap transferred in both. Both statements are made at the point
estimate. The question is whether the signed feature-response directions
1070 agree.

That check requires a labelled probe in the target region, and so, on
inspection, do the niche and regime diagnostics. Only the marginal family
can be computed before any target label exists, and it is the family that
failed. How large the probe has to be is not established here, so no label
1075 budget is claimed. A signed association with a usable interval may need
fewer labels than a refitted model, but that was not tested. What is clear is
that the supervised recalibration that label-free alignment cannot deliver is
not cheap. Section 4.9 prices it for three regions: thirty-two labelled 5 km
blocks recovered 85 to 89 % of the target ceiling in three of six directions, 51
1080 to 57 % in two more and 30 % in the sixth, and at its top budget the labelled
set already holds most of one target region’s burned cells. Where no target
labels exist, the transfer performance of such models should be treated as
unknown rather than inferred from similarity.

For model builders, local skill and portability should be reported sepa-
1085 rately rather than assumed to travel together. A predictor block worth a
large within-region increment may contribute nothing distinguishable from
zero across regions, with a sign that varies by pair, and reporting only within-
region validation hides that completely. Three extensions are left for fu-
ture work: temporal transfer, meteorological covariates, and physically nor-

malised dryness variables. Each is a step towards the self-calibrating thermal monitoring system that motivated this study.

CRedit authorship contribution statement

Yunus Emre Cogurcu: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Val- idation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All satellite inputs are public and are obtained through Google Earth Engine. The processing pipeline, its configuration and the frozen numeric outputs are publicly available; the repository, the commit of record and the licence are given in the data and code availability statement of the Methods, together with the three components that fall outside that release. No digital object identifier is minted and no archival deposit exists.

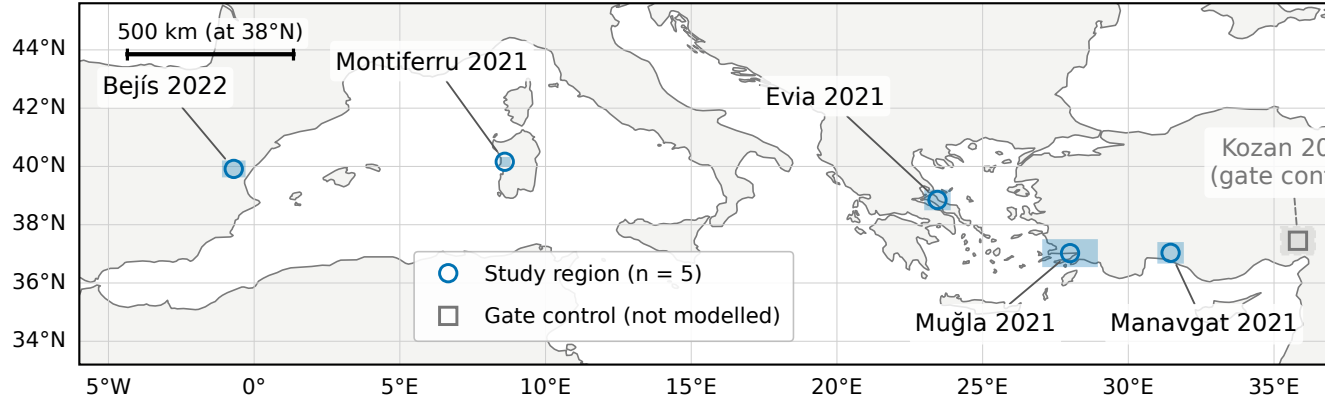


Figure 1: **Study regions and the gate negative control.** Locations of the five wildfire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

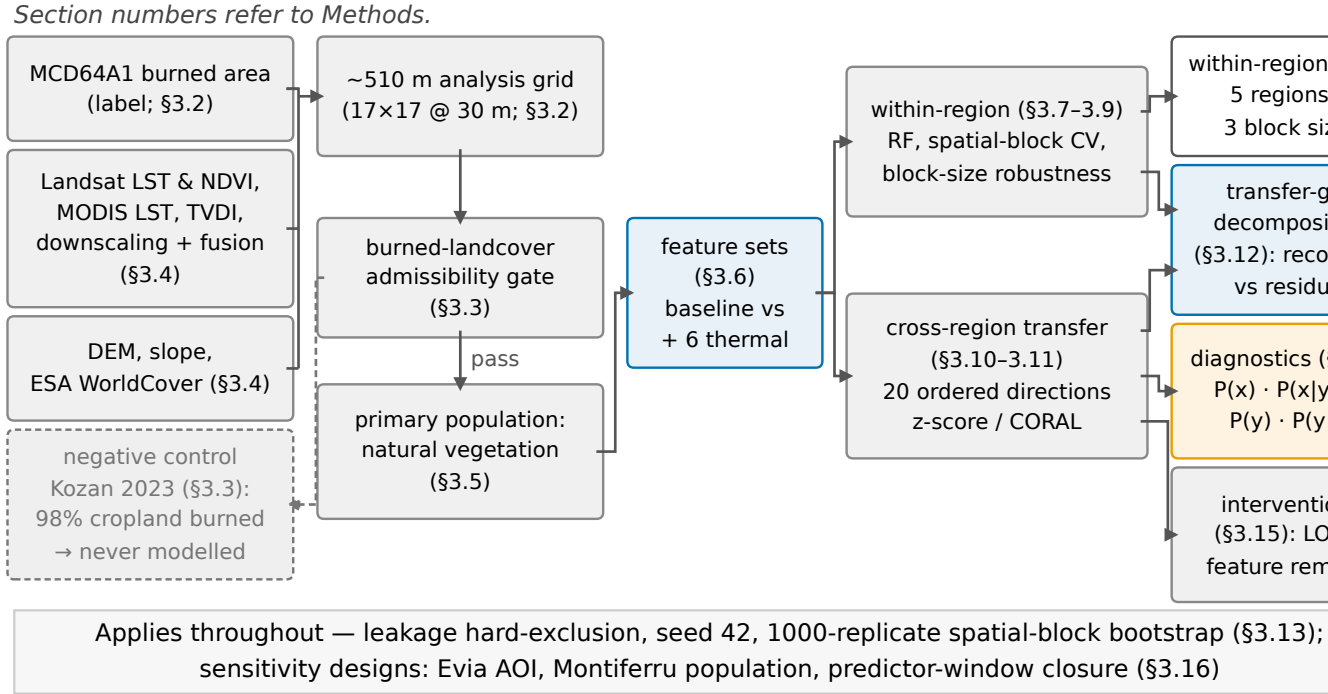


Figure 2: **Processing pipeline and evaluation programme.** Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

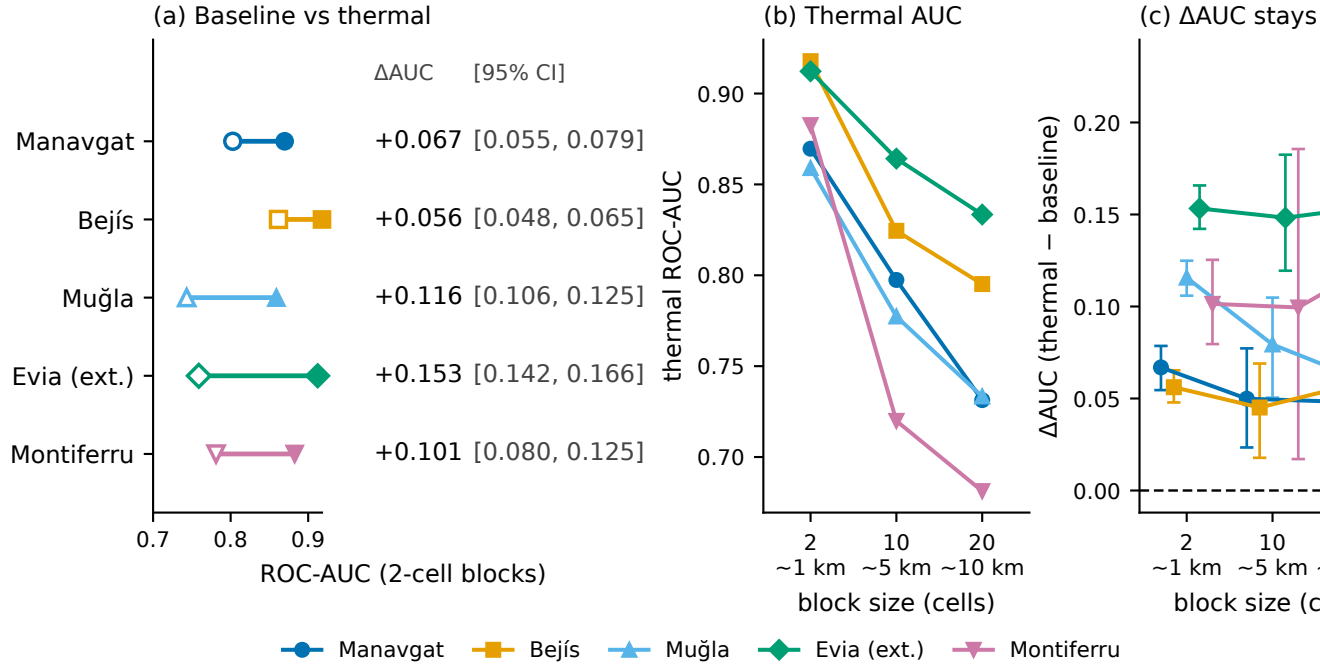


Figure 3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, ≈ 1 , 5 and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region–block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 2, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

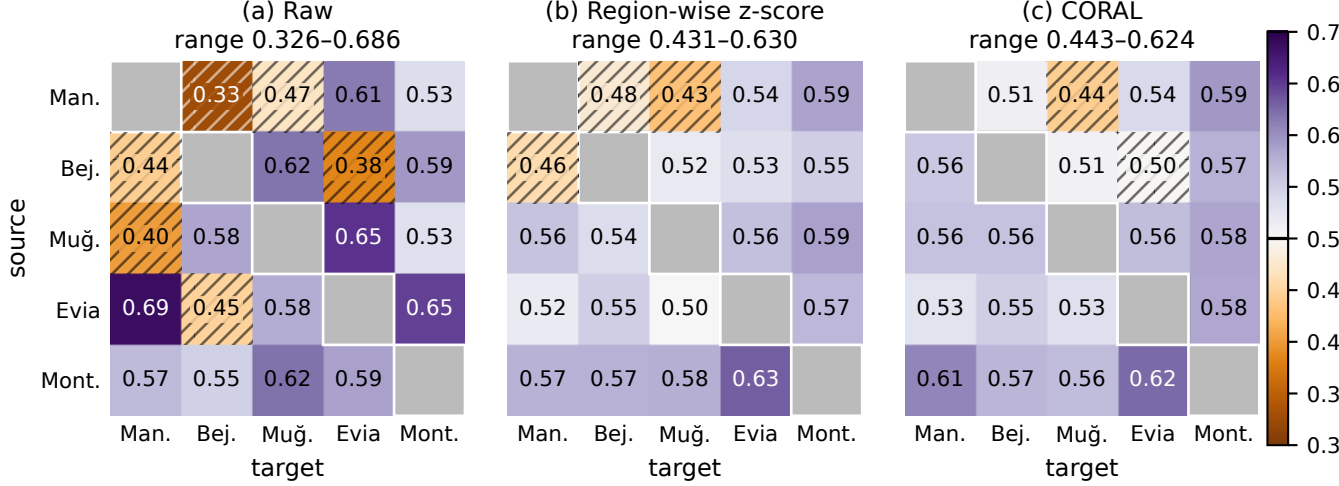


Figure 4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

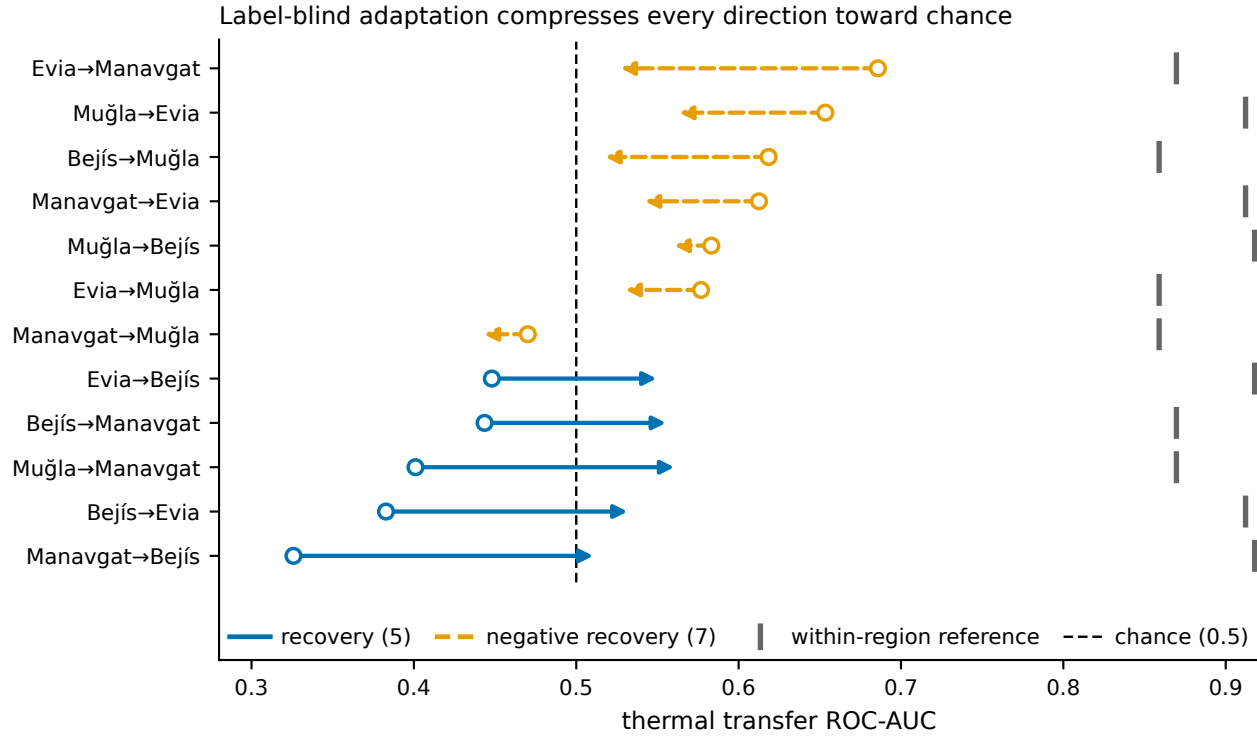


Figure 5: **Label-blind adaptation compresses transfer toward chance rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

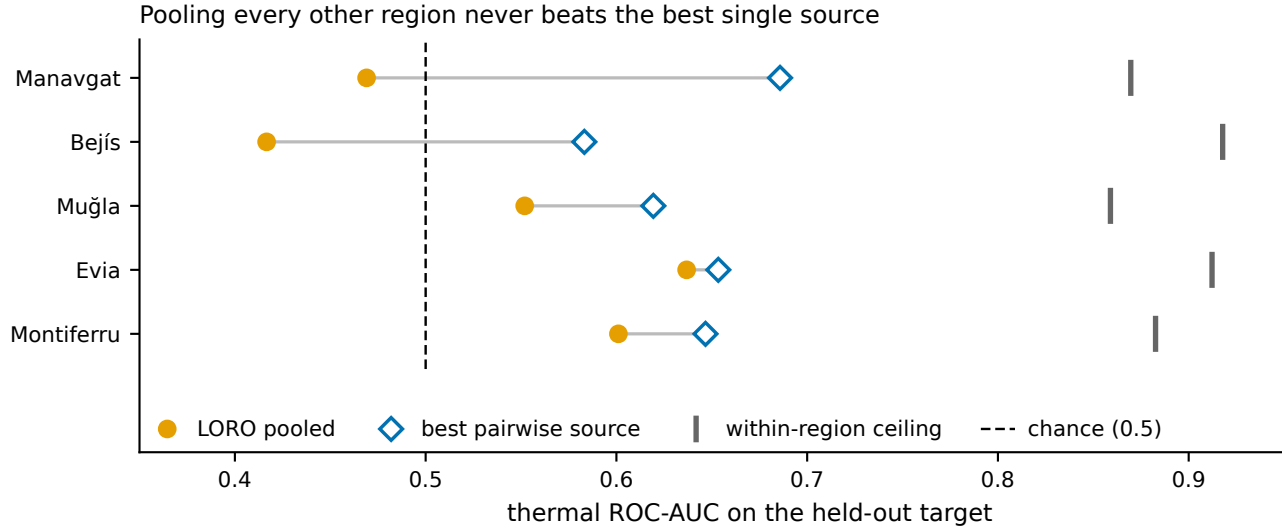


Figure 6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

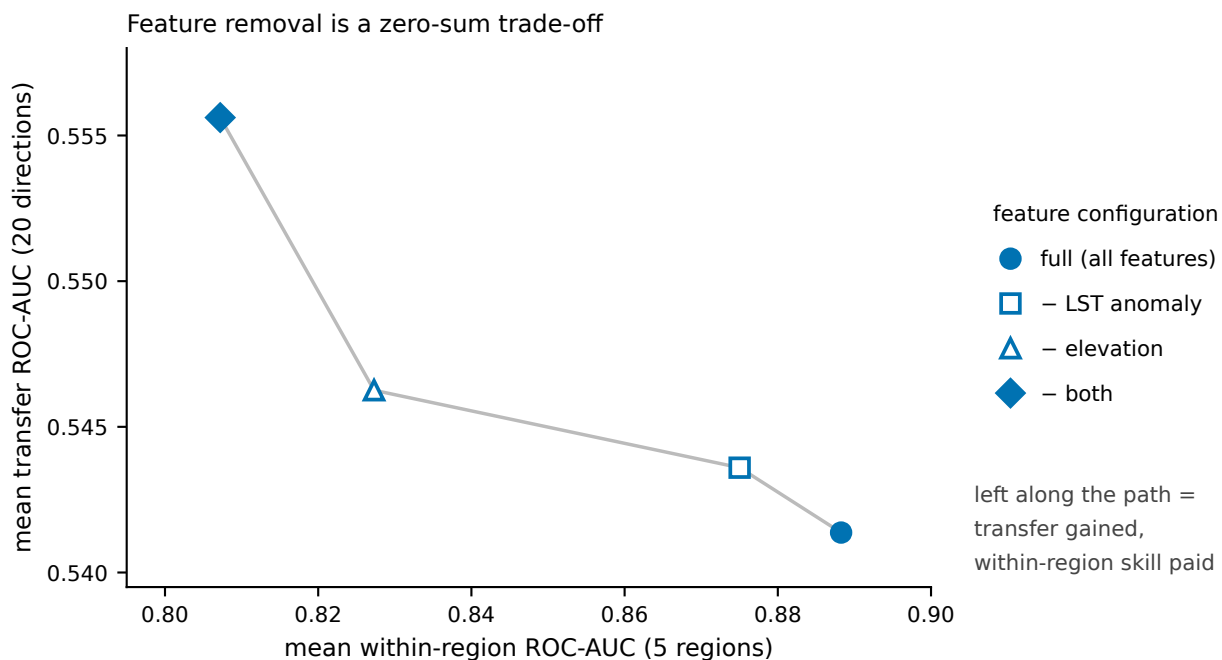


Figure 7: **Removing the direction-reversing features is a zero-sum trade-off.**

Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically (0.541 $\rightarrow$ 0.556) while within-region skill falls monotonically (0.888 $\rightarrow$ 0.807). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The transfer gain is 0.014 AUC and the within-region cost 0.081 AUC, about six times larger: what is removed is locally informative and non-portable, and removing it buys portability by giving up skill rather than by finding transferable signal.

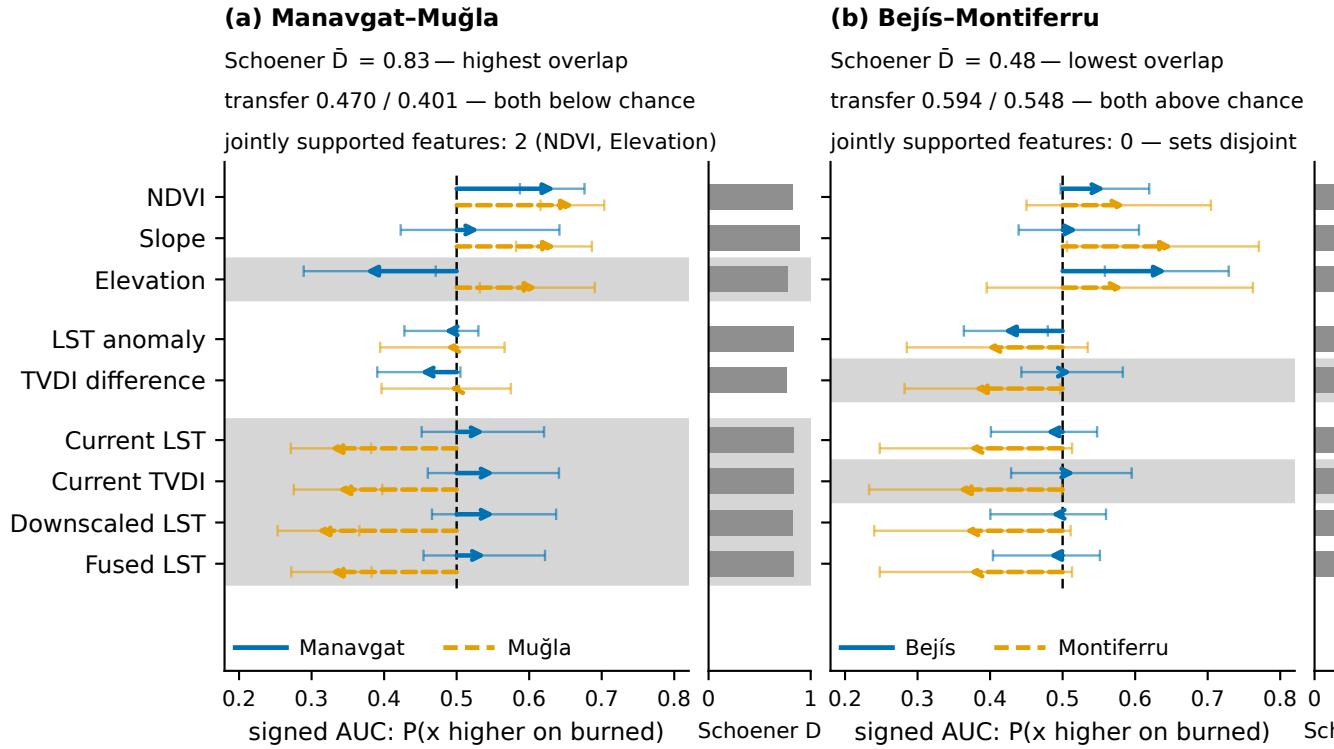


Figure 8: **Burned-niche overlap does not determine transfer: the contrast pair.**

Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat-Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís-Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

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